Yu-Lun Liu

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yulunalexliu.github.io

Research Interests

I work on *image/video processing*, *computer vision*, and *computational photography*, particularly on essential problems requiring *machine learning* with insights from *geometry* and *domain-specific knowledge*.

- Image and video synthesis and generation
- 3D from multi-view and sensors
- Low-level vision

Work Experience

Feb. 2026 – **Department of Computer Science, National Yang Ming Chiao Tung University**.
Associate Professor

Feb. 2023 – Feb. 2026 **Department of Computer Science, National Yang Ming Chiao Tung University**.
Assistant Professor

May 2022 – Oct. 2022 **Meta Reality Labs Research**.
Research Scientist Intern

Jun. 2017 – Feb. 2022 **MediaTek Inc**.
Senior Algorithm Development Engineer

Sep. 2014 – Jun. 2017 **MediaTek Inc**.
Algorithm Development Engineer

Selected Honors and Awards

Jan. 2026 **3rd place of the NYCU CS Undergraduate Research Competition**, *CS*, NYCU.
Dec. 2025 **Best Master Thesis Award**, *Taiwanese Association for Artificial Intelligence (TAAI)*.
Dec. 2025 **Excellent Master Thesis Award**, *Taiwanese Association for Artificial Intelligence (TAAI)*.
July 2025 **3rd place of the NYCU CS Undergraduate Research Competition**, *CS*, NYCU.
July 2025 **2nd place of the NYCU CS Undergraduate Research Competition**, *CS*, NYCU.
Mar. 2025 **K. T. Li Young Researcher Award**, *Institute of Information & Computing Machinery (IICM)*.
Dec. 2024 **Outstanding Contribution Award for Young CS Scholars**, *College of CS*, NYCU.
Nov. 2024 **Excellent Master Thesis Award**, *Taiwanese Association for Artificial Intelligence (TAAI)*.
Aug. 2024 **Excellent Master Thesis Award**, *The Chinese Image Processing and Pattern Recognition Society (IPPR)*.
July 2024 **3rd place of the NYCU CS Undergraduate Research Competition**, *CS*, NYCU.
July 2024 **2nd place of the NYCU CS Undergraduate Research Competition**, *CS*, NYCU.
June 2024 **NSTC 2030 Emerging Young Scholar**, *National Science and Technology Council*, R.O.C. (Taiwan).
May 2024 **CVPR 2024 Outstanding Reviewer**.
April 2024 **Google Research Scholar Award**.
Jan. 2024 **1st place of the NYCU CS Undergraduate Research Competition**, *CS*, NYCU.
Aug. 2023 **Google's Research Grant**, *Google University Relations Programs*.
Aug. 2023 **Best Ph.D. Thesis Award**, *The Chinese Image Processing and Pattern Recognition Society (IPPR)*.
Jul. 2023 **Yushan Young Fellow Program**, *Ministry of Education*, R.O.C. (Taiwan).
Nov. 2013 **Academic Achievement Award**, *Institute of Electronics*, NCTU.
May. 2013 **Academic Achievement Award**, *Institute of Electronics*, NCTU.

Education

2017–2022 **Ph.D.**, *National Taiwan University*, Taipei, Taiwan.
Department of Computer Science and Information Engineering, CMLAB
Thesis: Restoring and Enhancing Images and Videos by Combining Modeling and Learning
Advisor: Yung-Yu Chuang link

2012–2014 **M.S.**, *National Chiao-Tung University*, Hsinchu, Taiwan.
Institute of Electronics, CommLab
Thesis: Wide-Angle Virtual View Synthesis with Depth-Based Background Modeling
Advisor: Hsueh-Ming Hang link

2008–2012 **B.S.**, *National Chiao-Tung University*, Hsinchu, Taiwan.
Department of Electronics Engineering

Publications (Google Scholar profile)

Journal papers

IJCV 2024 **Portrait Distortion Correction with Perspective-Aware 3D GANs**.
Zhixiang Wang, Yu-Lun Liu, Jia-Bin Huang, Shin'ichi Satoh, Sizhuo Ma, Guru Krishnan, Jian Wang
arXiv, 2023 paper project

TPAMI 2021 **Learning to See Through Obstructions with Layered Decomposition**.
Yu-Lun Liu, Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
IEEE Transactions on Pattern Analysis and Machine Intelligence, 2021 paper project

Conference papers

CVPR 2026 **Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion**.
Ting-Hsuan Chen*, Ying-Huan Chen*, Tao Tu, Jie-Ying Lee, Cho-Ying Wu, Fangzhou Lin, Hengyuan Zhang, David Paz, Xinyu Huang, Yuliang Guo†, Yu-Lun Liu†, Yue Wang†, Liu Ren
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026

CVPR 2026 **Reflection Separation from a Single Image via Joint Latent Diffusion**.
Zheng-Hui Huang, Zhixiang Wang, Yu-Lun Liu, Yung-Yu Chuang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026

CVPR 2026 **PhaSR: Generalized Image Shadow Removal with Physically Aligned Priors**.
Chia-Ming Lee, Yu-Fan Lin, Yu-Jou Hsiao, Jin-Hui Jiang, Yu-Lun Liu, Chih-Chung Hsu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 paper project

CVPR 2026 **ReflexSplit: Single Image Reflection Separation via Layer Fusion-Separation**.
Chia-Ming Lee, Yu-Fan Lin, Jin-Hui Jiang, Yu-Jou Hsiao, Chih-Chung Hsu, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 paper project

CVPR 2026 **Voxify3D: Pixel Art Meets Volumetric Rendering**.
Yi-Chuan Huang, Jiewen Chan, Hao-Jen Chien, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026 paper project

WACV 2026 **PS3: Part level instance segmentation in 3D**.
Hong-Xuan Yen, Chiamin Chen, Yanqing Wang, Yu-Lun Liu, Min Sun
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026

WACV 2026 **HDR Reconstruction Boosting with Training-Free and Exposure-Consistent Diffusion**.
Yo-Tin Lin, Sykai Chen, Hou-Ning Hu, Yen-Yu Lin, Yu-Lun Liu
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026

WACV 2026 **TED-4DGS: Temporally Activated and Embedding-based Deformation for 4DGS Compression**.
Cheng-Yuan Ho, He-Bi Yang, Jui-Chiu Chiang, Yu-Lun Liu, Wen-Hsiao Peng
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 paper project

WACV 2026 **Splannequin: Freezing Monocular Mannequin-Challenge Footage with Dual-Detection Splatting**.
Hao-Jen Chien, Yi-Chuan Huang, Chung-Ho Wu, Wei-Lun Chao, Yu-Lun Liu
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2026 paper project

SIGGRAPH Asia 2025 **Prior-Enhanced Gaussian Splatting for Dynamic Scene Reconstruction from Casual Video**.
Meng-Li Shih, Ying-Huan Chen, Yu-Lun Liu, Brian Curless,
ACM SIGGRAPH Asia 2025 Conference Papers, 2025 paper project

NeurIPS 2025 **FIPER: Generalizable Factorized Fields for Joint Image Compression and Super-Resolution**.
Yang-Che Sun, Cheng Yu Yeo, Ernie Chu, Jun-Cheng Chen, Yu-Lun Liu
Conference on Neural Information Processing Systems, 2025 paper project

ICCV 2025 **LightsOut: Diffusion-based Outpainting for Enhanced Lens Flare Removal**.
Shr-Ruei Tsai, Wei-Cheng Chang, Jie-Ying Lee, Chih-Hai Su, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 paper project

ICCV 2025 **StealthAttack: Robust 3D Gaussian Splatting Poisoning via Density-Guided Illusions**.
Bo-Hsu Ke, You-Zhe Xie, Yu-Lun Liu, Wei-Chen Chiu
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 paper project

ICCV 2025 **LongSplat: Robust Unposed 3D Gaussian Splatting for Casual Long Videos**.
Chin-Yang Lin, Cheng Sun, Fu-En Yang, Min-Hung Chen, Yen-Yu Lin, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 paper project

ICCV 2025 **OpenM3D: Open Vocabulary Multi-view Indoor 3D Object Detection without Human Annotations**.
Peng-Hao Hsu, Ke Zhang, Fu-En Wang, Tao Tu, Ming-Feng Li, Yu-Lun Liu, Albert Y. C. Chen, Min Sun, Cheng-Hao Kuo
Proceedings of the IEEE/CVF Conference on Computer Vision, 2025 paper

CoRL 2025 **See, Point, Fly: A Learning-Free VLM Framework for Universal Unmanned Aerial Navigation**.
Chih Yao Hu, Yang-Sen Lin, Yuna Lee, Chih-Hai Su, Jie-Ying Lee, Shr-Ruei Tsai, Chin-Yang Lin, Kuan-Wen Chen, Tsung-Wei Ke, Yu-Lun Liu
9th Annual Conference on Robot Learning, 2025 paper project

CVPR 2025 **GCC: Generative Color Constancy via Diffusing a Color Checker**.
Chen-Wei Chang, Cheng-De Fan, Chia-Che Chang, Yi-Chen Lo, Yu-Chee Tseng, Jiun-Long Huang, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 paper project

CVPR 2025 **AuraFusion360: Augmented Unseen Region Alignment for Reference-based 360° Unbounded Scene Inpainting**.
Chung-Ho Wu*, Yang-Jung Chen*, Ying-Huan Chen, Jie-Ying Lee, Bo-Hsu Ke, Chun-Wei Tuan Mu, Yi-Chuan Huang, Chin-Yang Lin, Min-Hung Chen, Yen-Yu Lin, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 paper project

CVPR 2025 **SpectroMotion: Dynamic 3D Reconstruction of Specular Scenes**.
Cheng-De Fan, Chen-Wei Chang, Yi-Ruei Liu , Jie-Ying Lee, Jiun-Long Huang, Yu-Chee Tseng, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 paper project

CVPR 2025 **FrugalNeRF: Fast Convergence for Few-shot Novel View Synthesis without Learned Priors**.
Chin-Yang Lin*, Chung-Ho Wu*, Chang-Han Yeh, Shih-Han Yen, Cheng Sun, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 paper project

CVPR 2025 **DeNVeR: Deformable Neural Vessel Representations for Unsupervised Video Vessel Segmentation**.
Chun-Hung Wu, Shih-Hong Chen, Chih-Yao Hu, Hsin-Yu Wu, Kai-Hsin Chen, Yu-You Chen, Chih-Hai Su, Chih-Kuo Lee, Yu-Lun Liu
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025 paper project

ICLR 2025 **CAT-3DGS: A Context-Adaptive Triplane Approach to Rate-Distortion-Optimized 3DGS Compression**.
Yu-Ting Zhan*, Cheng-Yuan Ho*, Hebi Yang, Jui-Chiu Chiang, Yu-Lun Liu, Wen-Hsiao Peng
International Conference on Learning Representations, 2025 paper

ICASSP 2025 **3D Gaussian Splatting with Grouped Uncertainty for Unconstrained Images**.
Hao-Yu Hou*, Chia-Chi Hsu*, Yu-Chen Huang*, Mu-Yi Shen*, Wei-Fang Sun, Cheng Sun, Chia-Che Chang, Yu-Lun Liu, Chun-Yi Lee
IEEE International Conference on Acoustics, Speech, and Signal Processing, 2025 paper

WACV 2025 **ORFormer: Occlusion-Robust Transformer for Accurate Facial Landmark Detection**.
Jui-Che Chiang, Hou-Ning Hu, Bo-Syuan Hou, Chia-Yu Tseng, Yu-Lun Liu, Min-Hung Chen, Yen-Yu Lin
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 paper project
Oral Presentation

WACV 2025 **CorrFill: Enhancing Faithfulness in Reference-based Inpainting with Correspondence Guidance in Diffusion Models**.
Kuan-Hung Liu, Cheng-Kun Yang, Min-Hung Chen, Yu-Lun Liu, Yen-Yu Lin
Proceedings of IEEE/CVF Winter Conference on Applications of Computer Vision, 2025 paper project
Oral Presentation

NeurIPS 2024 **ReF-LDM: A Latent Diffusion Model for Reference-based Face Image Restoration**.
Chi-Wei Hsiao, Yu-Lun Liu, Cheng-Kun Yang, Sheng-Po Kuo, Yucheun Kevin Jou, Chia-Ping Chen
Conference on Neural Information Processing Systems, 2024 paper project

NeurIPS 2024 **Depth Anywhere: Enhancing 360 Monocular Depth Estimation via Perspective Distillation and Unlabeled Data Augmentation**.
Ning-Hsu Wang, Yu-Lun Liu
Conference on Neural Information Processing Systems, 2024 paper project

NeurIPS 2024 **NaRCan: Natural Refined Canonical Image with Integration of Diffusion Prior for Video Editing**.
Ting-Hsuan Chen, Jiewen Chan, Hau-Shiang Shiu, Shih-Han Yen, Chang-Han Yeh, Yu-Lun Liu
Conference on Neural Information Processing Systems, 2024 paper project

IROS 2024 **Precise Pick-and-Place using Score-Based Diffusion Networks**.
Shih-Wei Guo, Tsu-Ching Hsiao, Yu-Lun Liu, Chun-Yi Lee
IEEE/RSJ International Conference on Intelligent Robots and Systems, 2024 paper project

ECCV 2024 **GenRC: 3D Indoor Scene Generation from Sparse Image Collections**.
Ming-Feng Li, Yueh-Feng Ku, Hong-Xuan Yen, Chi Liu, Yu-Lun Liu, Albert Y. C. Chen, Cheng-Hao Kuo, Min Sun
Proceedings of European Conference on Computer Vision, 2024 paper project

ICRA Workshop 2024 **DriveEnv-NeRF: Exploration of A NeRF-Based Autonomous Driving Environment for Real-World Performance Validation**.
Mu-Yi Shen, Chia-Chi Hsu, Hao-Yu Hou, Yu-Chen Huang, Wei-Fang Sun, Chia-Che Chang, Yu-Lun Liu, Chun-Yi Lee
IEEE International Conference on Robotics and Automation RoboNerF Workshop, 2024 paper project

SIGGRAPH 2024 **Matting by Generation**.
Zhixiang Wang, Baiang Li, Jian Wang, Yu-Lun Liu, Jinwei Gu, Yung-Yu Chuang, Shin'ichi Satoh
ACM SIGGRAPH 2024 Conference Papers, 2024 paper project

SIGGRAPH 2024 **BoostMVSNeRFs: Boosting MVS-based NeRFs to Generalizable View Synthesis in Large-scale Scenes**.
Chih-Hai Su*, Chih-Yao Hu*, Shr-Ruei Tsai*, Jie-Ying Lee*, Chin-Yang Lin, Yu-Lun Liu
ACM SIGGRAPH 2024 Conference Papers, 2024 paper project

CVPR 2024 **Image-Text Co-Decomposition for Text-Supervised Semantic Segmentation**.
Ji-Jia Wu, Andy Chia-Hao Chang, Chieh-Yu Chuang, Chun-Pei Chen, Yu-Lun Liu, Min-Hung Chen, Hou-Ning Hu, Yung-Yu Chuang, Yen-Yu Lin
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024 paper

CVPR 2024 **HumanNeRF-SE: A Simple yet Effective Approach to Animate HumanNeRF with Diverse Poses**.
Caoyuan Ma, Yu-Lun Liu, Zhixiang Wang, Wu Liu, Xinchen Liu, Zheng Wang
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024 paper project

ICLR 2024 **Dual Associated Encoder for Face Restoration**.
Yu-Ju Tsai, Yu-Lun Liu, Lu Qi, Kelvin C.K. Chan, Ming-Hsuan Yang
The Twelfth International Conference on Learning Representations, 2024 paper project

AAAI 2024 **Improving Robustness for Joint Optimization of Camera Poses and Decomposed Low-Rank Tensorial Radiance Fields**.
Bo-Yu Cheng, Wei-Chen Chiu, Yu-Lun Liu
Proceedings of AAAI Conference on Artificial Intelligence, 2024 paper project

ICCV 2023 **Learning Continuous Exposure Value Representations for Single-Image HDR Reconstruction**.
Su-Kai Chen, Hung-Lin Yen, Yu-Lun Liu, Min-Hung Chen, Hou-Ning Hu, Wen-Hsiao Peng and Yen-Yu Lin
Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023 project

ICCV 2023 **ImGeoNet: Image-induced Geometry-aware Voxel Representation for Multi-view 3D Object Detection**.
Tao Tu, Shun-Po Chuang, Yu-Lun Liu, Cheng Sun, Ke Zhang, Donna Roy, Cheng-Hao Kuo, Min Sun
Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023 project

CVPR 2023 **Progressively Optimized Local Radiance Fields for Robust View Synthesis**.
Andreas Meuleman, Yu-Lun Liu, Chen Gao, Jia-Bin Huang, Changil Kim, Min H. Kim, Johannes Kopf
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023 paper project

CVPR 2023 **Robust Dynamic Radiance Fields**.
Yu-Lun Liu, Chen Gao, Andreas Meuleman, Hung-Yu Tseng, Ayush Saraf, Changil Kim, Yung-Yu Chuang, Johannes Kopf, Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023 paper project

ICLR 2022 **Denoising Likelihood Score Matching for Conditional Score-based Data Generation** .
Chen-Hao Chao, Wei-Fang Sun, Bo-Wun Cheng, Yi-Chen Lo, Chia-Che Chang, Yu-Lun Liu, Yu-Lin Chang, Chia-Ping Chen, and Chun-Yi Lee
International Conference on Learning Representations, 2022 paper

ICCV 2021 **Bridging Unsupervised and Supervised Depth from Focusvia All-in-Focus Supervision**.
Ning-Hsu Wang, Ren Wang, Yu-Lun Liu, Yu-Hao Huang, Yu-Lin Chang, Chia-Ping Chen, and Kevin Jou
Proceedings of IEEE/CVF Conference on Computer Vision, 2021 paper project

ICCV 2021 **Hybrid Neural Fusion for Full-frame Video Stabilization**.
Yu-Lun Liu, Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision, 2021 paper project

ICPR 2020 **Explorable Tone Mapping Operators**.
Chien-Chuan Su, Ren Wang, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Soo-Chang Pei
Proceedings of the 25th International Conference on Pattern Recognition, 2020 paper

ECCV 2020 **Learning Camera-Aware Noise Models**.
Ke-Chi Chang, Ren Wang, Hung-Jin Lin, Yu-Lun Liu, Chia-Ping Chen, Yu-Lin Chang, and Hwann-Tzong Chen
Proceedings of European Conference on Computer Vision, 2020 paper project

CVPR 2020 **Learning to See Through Obstructions**.
Yu-Lun Liu, Wei-Sheng Lai, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020 paper project

CVPR 2020 **Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline**.
Yu-Lun Liu*, Wei-Sheng Lai*, Yu-Sheng Chen, Yi-Lung Kao, Ming-Hsuan Yang, Yung-Yu Chuang, and Jia-Bin Huang
Proceedings of IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020 paper project

AAAI 2020 **Attention-based View Selection Networks for Light-field Disparity Estimation**.
Yu-Ju Tsai, Yu-Lun Liu, Yung-Yu Chuang, and Ming Ouhyoung
Proceedings of AAAI Conference on Artificial Intelligence, 2020 paper

AAAI 2019 **Deep Video Frame Interpolation using Cyclic Frame Generation**.
Yu-Lun Liu, Yi-Tung Liao, Yen-Yu Lin, and Yung-Yu Chuang
Proceedings of AAAI Conference on Artificial Intelligence, 2019 paper project
Oral Presentation

APSIPA 2014 **Background modeling using depth information**.
Yu-Lun Liu and Hsueh-Ming Hang
Signal and Information Processing Association Annual Summit and Conference, 2014

PCS 2013 **Virtual view synthesis using backward depth warping algorithm**.
Du-Hsiu Li, Hsueh-Ming Hang, and Yu-Lun Liu
Picture Coding Symposium, 2013

Preprints

arXiv 2024 **DiffIR2VR-Zero: Zero-Shot Video Restoration with Diffusion-based Image Restoration Models**.
Chang-Han Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, Ting-Hsuan Chen, Yu-Lun Liu
arXiv, 2024 paper project

Talks

Invited Seminar **Breaking Through 3DGS: Tackling Dynamics, Editing, Long Videos, and Security**.
Department of Electrical Engineering, NTHU, Hsinchu, Taiwan, Nov. 2025.

Guest Lecture **3D from Multi-view**.
MediaTek Inc., Hsinchu, Taiwan, July 2025.

Invited Talk **What School and Work Didn't Teach Me**.
Library, NYCU, Hsinchu, Taiwan, Sep. 2024.

Invited Seminar **Enhancing 3D Scene Reconstruction in NeRFs**.
CSIE, NTU, Taipei, Taiwan, Mar. 2024.

Keynote **AI Photographer: How Machines Are Changing the Way We See the World**.
Fun AI Winter Camp, NCKU, Tainan, Taiwan, Feb. 2024.

Invited Talk **Dynamic and Local Radiance Fields for Robust View Synthesis**.
MediaTek Inc., Hsinchu, Taiwan, May. 2023.

Invited Talk	**Dynamic and Local Radiance Fields for Robust View Synthesis**. Institute of Communications Engineering, NTHU, Hsinchu, Taiwan, April. 2023.
Invited Talk	**Dynamic and Local Radiance Fields for Robust View Synthesis**. The 5th Augmented Intelligence and Interaction (AII) Workshop, Taoyuan, Taiwan, March. 2023.
Guest Lecture	**What School and the Workplace Didn't Teach Me**. NYCU, Hsinchu, Taiwan, March. 2023.
Invited Talk	**Learning to See the Unseen**. Taichung Municipal Taichung First Senior High School, Taichung, Taiwan, March. 2023.
Invited Talk	**Learning to See the Unseen**. Research Center for Information Technology Innovation, Academia Sinica, Taipei, Taiwan, January. 2023.
Invited Talk	**Learning to See the Unseen**. Institute of Information Systems and Applications, NTHU, Hsinchu, Taiwan, November. 2022.
Guest Lecture	**Recent Progress on Video Restoration**. CommE7005, NTU, Taipei, Taiwan, Mar. 2022.
Guest Lecture	**Rethinking Multimedia and IC Design in the Era of AI**. CommE7005, NTU, Taipei, Taiwan, Nov. 2021.
Guest Lecture	**CV Final Project - Video Frame Interpolation**. EEE5053, NTU, Taipei, Taiwan, May. 2021.
Guest Lecture	**Rethinking Multimedia and IC Design in the Era of AI**. DEE4592, NCTU, Hsinchu, Taiwan, Mar. 2021.
Invited Talk	**Learning to See Through Obstructions**. 2020 CVPR Taiwan Sharing Session, NTU, Taipei, Taiwan, July. 2020.
Invited Talk	**Single-Image HDR Reconstruction by Learning to Reverse the Camera Pipeline**. 2020 CVPR Taiwan Sharing Session, NTU, Taipei, Taiwan, July. 2020.
Guest Lecture	**CV Final Project - Depth Estimation**. EEE5053, NTU, Taipei, Taiwan, Jan. 2020.
Invited Talk	**Deep Video Frame Interpolation using Cyclic Frame Generation**. The 3rd Augmented Intelligence and Interaction (AII) Workshop, Taoyuan, Taiwan, July. 2019.
Oral	**Deep Video Frame Interpolation using Cyclic Frame Generation**. AAAI, Honolulu, Hawaii, USA, Jan. 2019.
Guest Lecture	**CV Final Project - Depth Estimation**. EEE5053, NTU, Taipei, Taiwan, Jan. 2019.

Academic Services

Publication Chair	ACCV 2024
COI Coordinator	SIGGRAPH Asia 2025
Area Chair	NeurIPS 2025 MVA 2025
International Program Committee	Pacific Graphics 2025
Organizer	2020 CVPR Taiwan Sharing Session
Thesis Committees	Hsuan-Kung Yang (advisor: Chun-Yi Lee) - PhD, expected 2024 Yi-Hsin Chen (advisor: Wen-Hsiao Peng) – PhD, expected 2024 Ardian Umam (advisors: Yen-Yu Lin and Jen-Hui Chuang) – PhD, Dec 2024 Cheng Sun (advisors: Hwann-Tzong Chen and Min Sun) – PhD, Jul 2023 Hitika Tiwari (advisors: K. S. Venkatesh and Yong-Sheng Chen) – PhD, Jul 2023 Cheng-Kun Yang (advisors: Yung-Yu Chuang and Yen Yu-Lin) – PhD, Jul 2023

Journal Reviewer	IEEE Transactions on Image Processing (**TIP**) ACM Computing Surveys (**CSUR**) Applied Soft Computing Pattern Recognition ACM Transactions on Graphics (**TOG**) Information Fusion IEEE Transactions on Multimedia International Journal of Computer Vision The Journal of Supercomputing Computer Vision and Image Understanding Computers & Graphics Multidimensional Systems and Signal Processing International Journal of Pattern Recognition and Artificial Intelligence
Conference Reviewer	IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022, 2023, 2024, 2025 IEEE Conference on Computer Vision (**ICCV**), 2021, 2023, 2025 European Conference on Computer Vision (**ECCV**), 2020, 2022, 2024 British Machine Vision Conference (**BMVC**), 2023, 2024 Asian Conference on Computer Vision (**ACCV**), 2020, 2022, 2024 Conference on Neural Information Processing Systems (**NeurIPS**), 2022, 2023, 2024 International Conference on Learning Representations (**ICLR**), 2022, 2023, 2024, 2025 International Conference on Machine Learning (**ICML**), 2023, 2024 ACM SIGGRAPH (**SIGGRAPH**), 2022, 2023, 2024, 2025 ACM SIGGRAPH Asia (**SIGGRAPH**), 2024, 2025 AAAI Conference on Artificial Intelligence (**AAAI**), 2021, 2022, 2023, 2024, 2025 International Joint Conference on Artificial Intelligence (**IJCAI**), 2021, 2022, 2023, 2024, 2025 ACM International Conference on Multimedia (**MM**), 2018 IEEE International Conference on Multimedia and Expo Workshops (**ICMEW**), 2023 International Conference on Robotics and Automation (**ICRA**), 2023 Conference on Robot Learning (**CoRL**), 2025
Research Mentor	Research Mentor for interns at MediaTek Inc., 2019, 2020, 2021, 2022.
Professional Societies	IEEE student member AAAI student member TAAI member IPPR member

Teaching Experiences

Fall 2025	**CSIC30191 Image and Video Generations**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Fall 2025	**CSIC30107 Video Compression**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Spring 2025	**CSCS10017 Signals and Systems**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Fall 2024	**CSIC30107 Video Compression**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Spring 2024	**CSCS10017 Signals and Systems**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Fall 2023	**CSIC30107 Video Compression**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Spring 2023	**CSIC30107 Video Compression**, *Department of Computer Science, National Yang Ming Chiao Tung University*.
Spring 2013	**Teaching Assistant of DEE1315 Probability and Statistics with Prof. Hsueh-Ming Hang**, *Department of Electronics Engineering, National Chiao Tung University*.

References

Ph.D. Advisor	**Yung-Yu Chuang**, *Professor of Department of Computer Science and Information Engineering*, National Taiwan University. ✉ cyy@csie.ntu.edu.tw ℹ homepage
M.S. Advisor	**Hsueh-Ming Hang**, *Professor of Department of Electronics Engineering*, National Chiao Tung University. ✉ hmhang@mail.nctu.edu.tw ℹ homepage
Director	**Johannes Kopf**, *Director of the Computational Photography Group*, Meta. ✉ johannes.peter.kopf@gmail.com
Director	**Shaw-Min Lei**, *Director of the Multimedia Technology Development Division*, MediaTek Inc. ✉ shawmin.lei@mediatek.com
Manager of Department	**Yu-Lin Chang**, *Manager of the Intelligent Vision Processing Department*, MediaTek Inc. ✉ yulinchang@google.com
Manager of Department	**Chia-Ping Chen**, *Manager of the Intelligent Vision Processing Department*, MediaTek Inc. ✉ chiaping.chen@mediatek.com
Research Mentor	**Ming-Hsuan Yang**, *Professor of Electrical Engineering and Computer Science*, University of California at Merced. ✉ mhyang@ucmerced.edu ℹ homepage
Research Mentor	**Jia-Bin Huang**, *Associate Professor of Department of Computer Science*, University of Maryland, College Park. ✉ jbhuang@umd.edu ℹ homepage
Research Mentor	**Yen-Yu Lin**, *Professor of Department of Computer Science*, National Yang Ming Chiao Tung University. ✉ lin@cs.nctu.edu.tw ℹ homepage
Research Mentor	**Wei-Sheng Lai**, *Software Engineer*, Google. ✉ wlai24@ucmerced.edu ℹ homepage
Research Mentor	**Changil Kim**, *Research Scientist*, Meta. ✉ changil@csail.mit.edu ℹ homepage
Research Mentor	**Hung-Yu Tseng**, *Research Scientist*, Meta. ✉ hytseng0509@gmail.com ℹ homepage
Research Mentor	**Chen Gao**, *Research Scientist*, Meta. ✉ gaochen@meta.com ℹ homepage